Ejercicios de tcpdump

- **Ejercicios básicos**

1. **Listar interfaces disponibles**

tcpdump -D

```
osboxes@osboxes:~$ tcpdump -D
1.enp0s3 [Up, Running, Connected]
2.any (Pseudo-device that captures on all interfaces) [Up, Running]
3.lo [Up, Running, Loopback]
4.bluetooth-monitor (Bluetooth Linux Monitor) [Wireless]
5.nflog (Linux netfilter log (NFLOG) interface) [none]
6.nfqueue (Linux netfilter queue (NFQUEUE) interface) [none]
7.dbus-system (D-Bus system bus) [none]
8.dbus-session (D-Bus session bus) [none]
osboxes@osboxes:~$
```

👉 Muestra todas las interfaces de red que tcpdump puede usar.

2. **Capturar tráfico en la interfaz eth0**

tcpdump -i eth0

```
osboxes@osboxes:~$ sudo tcpdump -i enp0s3
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
03:57:22.424080 ARP, Request who-has 10.0.2.2 tell 10.0.2.15, length 46
03:57:22.723145 ARP, Request who-has 10.0.2.2 tell 10.0.2.15, length 28
03:57:23.266666 IP 192.168.0.99.56825 > 192.168.1.255.2000: UDP, length 4
03:57:23.452550 ARP, Request who-has 10.0.2.2 tell 10.0.2.15, length 46
03:57:23.452557 ARP, Request who-has 10.0.2.15 tell 10.0.2.15, length 46
03:57:23.452971 ARP, Request who-has 10.0.2.15 tell 10.0.2.15, length 46
03:57:23.747143 ARP, Request who-has 10.0.2.2 tell 10.0.2.15, length 28
```

👉 Ver los primeros 20 paquetes que pasan por eth0.

3. **Guardar captura en un fichero**

tcpdump -i eth0 -w trafico.log

```
osboxes@osboxes:~$ sudo tcpdump -i enp0s3 -w
tcpdump: option requires an argument -- 'w'
tcpdump version 4.99.4
libpcap version 1.10.5 (with TPACKET_V3)
OpenSSL 3.4.1 11 Feb 2025
Usage: tcpdump [-AbdDefhHIJKlLnNOpqStuUvxX#] [ -B size ] [ -c count ] [--count]
                [ -C file_size ] [ -E algo:secret ] [ -F file ] [ -G seconds ]
                [ -i interface ] [ --immediate-mode ] [ -j tstamptype ]
                [ -M secret ] [ --number ] [ --print ] [ -Q in|out|inout ]
                [ -r file ] [ -s snaplen ] [ -T type ] [ --version ]
                [ -V file ] [ -w file ] [ -W filecount ] [ -y datalinktype ]
                [ --time-stamp-precision precision ] [ --micro ] [ --nano ]
                [ -z postrotate-command ] [ -Z user ] [ expression ]
osboxes@osboxes:~$
```

👉 Luego ábrelo con:

tcpdump -r trafico.log

```
osboxes@osboxes:~$ sudo tcpdump -r trafico.log
tcpdump: trafico.log: No such file or directory
osboxes@osboxes:~$
```

- **Filtrado por protocolos y puertos**

4. **Capturar solo tráfico TCP en eth0**

tcpdump -i eth0 tcp

```
osboxes@osboxes:~$ sudo tcpdump -i enp0s3 tcp
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
```

5. **Capturar tráfico en el puerto 80 (HTTP)**

tcpdump -i eth0 tcp port 80

```
osboxes@osboxes:~$ sudo tcpdump -i enp0s3 tcp port 80
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
^C
0 packets captured
0 packets received by filter
0 packets dropped by kernel
```

6. **Capturar tráfico DNS (UDP puerto 53)**

tcpdump -i eth0 udp port 53

```
osboxes@osboxes:~$ sudo tcpdump -i enp0s3 udp port 53
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
```

- **Filtrado por IP**

7. **Tráfico desde una IP concreta (host origen)**

tcpdump -i eth0 src 192.168.43.143

```
osboxes@osboxes:~$ sudo tcpdump -i enp0s3 src 10.0.2.15
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
04:14:45.731963 ARP, Request who-has 10.0.2.2 tell 10.0.2.15, length 28
04:14:46.755917 ARP, Request who-has 10.0.2.2 tell 10.0.2.15, length 28
04:14:47.779159 ARP, Request who-has 10.0.2.2 tell 10.0.2.15, length 28
04:14:48.803483 ARP, Request who-has 10.0.2.2 tell 10.0.2.15, length 28
04:14:49.827144 ARP, Request who-has 10.0.2.2 tell 10.0.2.15, length 28
04:14:50.851196 ARP, Request who-has 10.0.2.2 tell 10.0.2.15, length 28
```

8. **Tráfico hacia una IP concreta (host destino)**

tcpdump -i eth0 dst 192.168.43.143

```
osboxes@osboxes:~$ sudo tcpdump -i enp0s3 dst 10.0.2.15
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
04:16:44.141856 ARP, Request who-has 10.0.2.15 tell 10.0.2.15, length 28
04:16:44.142060 ARP, Request who-has 10.0.2.15 tell 10.0.2.15, length 46
04:16:44.142907 ARP, Request who-has 10.0.2.15 tell 10.0.2.15, length 46
04:16:44.142914 ARP, Request who-has 10.0.2.15 tell 10.0.2.15, length 46
04:16:44.142914 ARP, Request who-has 10.0.2.15 tell 10.0.2.15, length 46
04:16:44.146458 ARP, Request who-has 10.0.2.15 tell 10.0.2.15, length 46
04:16:54.372069 ARP, Request who-has 10.0.2.15 tell 10.0.2.15, length 46
04:16:54.372201 ARP, Request who-has 10.0.2.15 tell 10.0.2.15, length 28
04:16:54.372642 ARP, Request who-has 10.0.2.15 tell 10.0.2.15, length 46
04:16:54.372756 ARP, Request who-has 10.0.2.15 tell 10.0.2.15, length 46
```

9. **Tráfico de y hacia una IP concreta (host completo)**

tcpdump host 192.168.1.143

```
osboxes@osboxes:~$ sudo tcpdump host 10.0.2.15
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
04:17:59.979099 ARP, Request who-has 10.0.2.2 tell 10.0.2.15, length 46
04:17:59.987777 ARP, Request who-has 10.0.2.2 tell 10.0.2.15, length 28
04:18:00.307334 IP 10.0.2.15.mdns > 224.0.0.251.mdns: 0*- [0q] 1/0/0 (Cache flus
h) PTR osboxes-3.local. (113)
04:18:00.995145 ARP, Request who-has 10.0.2.2 tell 10.0.2.15, length 28
04:18:01.002779 ARP, Request who-has 10.0.2.2 tell 10.0.2.15, length 46
04:18:02.019145 ARP, Request who-has 10.0.2.2 tell 10.0.2.15, length 28
```

- **Filtrado por MAC y red**

10. **Tráfico con destino a una dirección MAC**

tcpdump ether dst 8A:B1:11:A0:BC:53

```
osboxes@osboxes:~$ sudo tcpdump ether dst 8A:B1:11:A0:BC:53
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
```

11. **Tráfico de una red completa**

tcpdump net 192.168.43.0/24

```
osboxes@osboxes:~$ sudo tcpdump net 10.0.2.0/24
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
04:23:22.381472 ARP, Request who-has 10.0.2.2 tell 10.0.2.15, length 46
04:23:22.531129 ARP, Request who-has 10.0.2.2 tell 10.0.2.15, length 28
04:23:23.405091 ARP, Request who-has 10.0.2.2 tell 10.0.2.15, length 46
04:23:23.405233 ARP, Request who-has 10.0.2.15 tell 10.0.2.15, length 28
04:23:23.405485 ARP, Request who-has 10.0.2.15 tell 10.0.2.15, length 46
04:23:23.405490 ARP, Request who-has 10.0.2.15 tell 10.0.2.15, length 46
```

- **Visualización de datos**

12. **Mostrar el contenido en ASCII**

tcpdump -i eth0 -A

```
osboxes@osboxes:~$ sudo tcpdump -i enp0s3 -A
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
```

13. **Mostrar el contenido en hexadecimal**

tcpdump -i eth0 -XX

```
osboxes@osboxes:~$ sudo tcpdump -i enp0s3 -XX
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
```

🔹 **Ejercicios combinados (operadores lógicos)**

14. **Tráfico desde una IP y en puerto 80**

tcpdump -i eth0 src 192.168.43.143 and tcp port 80

```
osboxes@osboxes:~$ sudo tcpdump -i enp0s3 src 192.168.0.1 and tcp port 80
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
```

15. **Tráfico desde una IP excluyendo UDP**

tcpdump -i eth0 src 192.168.1.143 and not udp

```
osboxes@osboxes:~$ sudo tcpdump -i enp0s3 src 192.168.0.1 and not udp
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
```

16. **Tráfico desde una IP en HTTP o HTTPS**

tcpdump -i eth0 src 192.168.1.143 and (port http or https)

```
osboxes@osboxes:~$ sudo tcpdump -i enp0s3 src 192.168.0.1 and port http
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
^[[A^C
0 packets captured
0 packets received by filter
0 packets dropped by kernel
osboxes@osboxes:~$ sudo tcpdump -i enp0s3 src 192.168.0.1 and port http or https
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
```

👉 Con esta lista de **16 ejercicios** tienes un recorrido completo:

- Empezando por capturar y guardar tráfico.
- Luego filtrando por protocolo, puerto, IP, MAC y red.
- Finalmente usando operadores lógicos y visualización en distintos formatos.